

PPA Scaling of Flip FET Technology Down to A2 Node Enabled by Architecture Innovations: Self-aligned Gate, 2T Design with Embedded Power Rail and Ultra-stacked 4-Tier Transistors

Wanyue Peng†, Haoran Lu†, Jingru Jiang, Rui Guo, Jiacheng Sun, Jianxiang Jin, Yuji Cheng, Shengcheng Zhou, Ziqiao Xu, Chuan Lan, Yanbang Chu, Xun Jiang, Feiyu Teng, Ming Li, Yibo Lin, Xinwei Wang, Runsheng Wang, Heng Wu*, Ru Huang

†These authors contributed equally, School of Integrated Circuits, Peking University, Beijing, China, *Email: hengwu@pku.edu.cn

Abstract: In this work, we carried out a critical examination of Flip FET (FFET) technology from process innovation to circuit design for nodes from A14 to A2. The Fully-aligned FFET (F3ET) featuring self-aligned (SA) FS and BS gates was experimentally demonstrated, with common gate (CG) & split gate (SG) on fins and multi-stack Nanosheets. The Forksheets-based F3ET (F4ET) with embedded Power Rail (ePR) was proposed to reduce the cell height (CH) to 2T. With ultra-stacked 4-tier transistors, the Complementary FET (CFET) based FFET (CFFET) shows further area scaling potential. Comprehensive Power-Performance-Area (PPA) evaluation was conducted on both circuit- and block-level. DTCO knobs were also carefully studied on the A7 F3ET_{NS} and A5 F4ET and improved the RO frequency (HP) by 11.3% and 6.2% at iso-leakage respectively. A3 HP F4ET outperforms A14 HP Fin-based FFET (FFET_{Fin}) by 38.9% at iso-power @ $V_{dd} = 0.7$ V. P&R results on 32-bit RISC-V cores shows 44.9% (HP) / 49.8% (HD) area scaling and 20.0% (HP) / 27.9% (HD) frequency improvement from A14 to A5. SRAM scaling down to A2 was also studied in a 256×256 array.

Introduction: With the ending of pitch scaling, the advanced logic development follows two trends: 3D transistor stacking [1-4] and backside interconnects [4-7]. We proposed the FFET [8] previously, combining stacked transistors and symmetric FS-BS signal & power for the first time, with clear parasitics and area benefits demonstrated [9-10]. In this work, we further proposed several crucial process innovations to drive the continuous scaling of FFET from A14 to A2 for both high performance (HP) and high density (HD) applications, as on the roadmap shown in Fig. 1. To solve the issue of misaligned frontside (FS) and backside (BS) gates [11], we introduced and experimentally demonstrated the F3ET with the critical SA gate and its performance gain was studied at A7. For A5 & A3, we combined the F3ET and Forksheet [12] with the ePR into F4ET, pushing the CH to 2T. Circuit- and block-level PPA evaluation validated the scalability of FFET down to A3. Moreover, CFFET, a new architecture with ultra-stacked 4-tier transistors [13] enabled by the unique dual-sided process of FFET, was introduced at A2. At last, the scaling potential of the FFET SRAM down to A2 was also studied.

Device Fabrication of the F3ET: Fig. 2(a) shows the key steps of the Fin-based F3ET flow. The S/D placeholder (PH) and FS S/D are formed in sequence after the SA active and SA DG, followed by the 1st Flip. Then, BS dummy gate (DG) is stripped after BS S/D epi, followed by the CG & SG patterning. After the dual-sided HK & TiN are deposited together, the BS RMG and BEOL are formed, followed by the 2nd Flip [14]. At last, the FS RMG and other BEOL processes are finished. Note that the N/P S/D epi formed separately on each side (the key difference w.r.t the CFET) has no high aspect ratio (AR) challenges as in CFET [8] and the dual-sided metal gate is gate-last without thermal budget concerns. Fig. 2(b) shows the TEMs of the SG (left, oxide on the FS) & CG (right) in F3ET_{Fin}. Fig. 2(c) give the SEM & TEM of the multi-stack Nanosheets with partial channel release, proving the Nanosheet-based F3ET (F3ET_{NS}) (Fig. 2(d)).

TCAD and Compact Model: We assumed the nFET/pFET on the FS/BS. Process assumptions of A14 FFET_{Fin}, A10 Nanosheet-based FFET (FFET_{NS}), A7 F3ET_{NS}, A5 & A3 F4ET and A2 CFFET are listed in Table 1 & 2. HP cells have larger CH and stronger current drivability than HD cells. All HP devices used for the ring oscillator (RO) simulation and library characterization have the same I_{off} of 2 nA while for HD devices the I_{off} is 20 pA [15]. BSIM-CMG [16] compact models (CMs) were fitted to TCAD I_dV_g (Fig. 3), I_dV_d and CV curves (not shown).

RO PP roadmaps towards A3: A 15-stage INVD1-based RO with fan-out (FO) of 3 and distributed BEOL loads was used to compare the circuit-level performance from A14 to A3. The BEOL loads were extracted from P&R critical paths (CPs) [14, 17] of 32-bit RISC-V cores, as discussed later in Fig. 8(c). Power-frequency curves of HP FFETs in Fig. 4(a) shows that A10, A7, A5 and A3 HP FFETs outperform A14 HP FFET_{Fin} by 11.4%, 24.8%, 34.1% and 38.9% at iso-power @ $V_{dd} = 0.7$ V w/o BEOL load. Fig. 4(b) compares the I_{eff} vs C_{eff} relation of both HP and HD FFETs with various BEOL loads at iso-leakage @ $V_{dd} = 0.7$ V. A5 HP/HD F4ET outperforms A14 HP/HD FFET in speed by 85.6%/85.2% w/ short load. A3 and A5 FFET share the similar frequency while A3 FFET shows less power (-13.5% for HP and -24.2% for HD in average) due to the reduced C_{eff} .

F3ET DTCO: Besides the area benefits free from FS/BS gates misalignment, the SA gate in A7 F3ET_{NS} also has great benefits in performance. Fig. 5 (a) shows the F3ET_{NS} has much wider gate connections compared to the FFET w/o SA gate at the same sheet width (W_{NS}) and thus smaller gate resistance (R_{gate}). Besides, the W_{NS} in F3ET_{NS} can be further enlarged thanks to the elimination of gate merge vias [8]. These gives clear benefits in circuits with multistage cascade. For instance, for the AND2D1 in Fig. 5(b), design v1 cascades the independent INVD1 and NAND2D1. The other more area-efficient design (v2) uses the CG of the INVD1 to form the NAND2D1, as in Fig. 5(c). Fig. 5(d) shows the circuit speed is affected more by INVD1's R_{gate} for the v2 design and the RO for evaluating the two designs. Fig. 5(e) shows the RO frequency of the v2 design is 2.4% higher than v1. Furthermore, thanks to the SA gate in F3ET_{NS}, the R_{gate} is reduced by 88.7% at iso- W_{NS} , gaining an extra 1.6% frequency. Also, F3ET_{NS} has larger W_{NS} and thus 7.2% higher frequency (+ 11.3% w.r.t v1) with the R_{gate} considerably small.

F4ET DTCO: For further CH shrinking, Forksheet is further introduced into A5 FFET while keeping the active footprint [12]. Also, power rails at the cell boundary (Fig. 1) which limit the CH scaling are embedded into the dielectric wall of the Forksheet (Fig. 6(a)). This helps reduce the CH from 2.5T [8] to 2T in A5 HD F4ET. BS contact (BSC) (Fig. 6(b)) is used to connect the S/D epi and the ePR. BSC optimization such as the low temperature (LT) regrowth of heavily-doped epi [18], replacing BSC metals [18] and using wrapped around contact (WAC) [4] (Fig. 6(b)) help reduce the N/P source resistance (R_{source}) by 43.9%/27.4% and improve the RO frequency by 6.2% (Fig. 6(c)). Performance degradation in A5 HP F4ET caused by the IR-drop on the ePR resistance (R_{ePR}) was also evaluated by the RO in Fig. 6(d). Switching the ePR metal from W to Ru [19] and increasing ePR thickness can reduce the R_{ePR} by 68.9% and the frequency degradation w.r.t ignoring IR-drop can be limited to 0.8% with worst-case IR-drop ~ 16 mV (Figs. 6(e-f)).

FFET vs CFET W/ BSC: Like FFET, CFET with BSC has been proposed recently [2, 20-22], with better design flexibility [23]. As shown in Fig. 7 (a), CFET w/o BSC (v1) and two types of CFETs w/ BSC (CFET w/ TSMV & SA BSC [3-4] (v2.1) and CFET w/ symmetric FS-BS interconnects (v2.2) [22]) are compared with FFET at the same CH at A7. The FFET shows area gain over all CFETs in cells with SG [8] (Fig. 7(b)). HD F3ET_{NS} and HD CFET v2.2 share similar RO performance and outperform other two CFETs (Fig. 7(c)) for larger W_{NS} (Fig. 7(a)) and reduced parasitics thanks to the dual-sided power [3, 8-10]. The HP F3ET_{NS} gains an extra 7.3% higher frequency at iso-leakage for its smaller gate extension compared to the HP CFET v2.2 without BS COAG [22] (Fig. 7(d)). Furthermore, with input pins on the BS, CFET v2.2 shows worse performance (Fig. 7(c)) due to the larger BS parasitics from the remained BS sub-fin Si while F3ET_{NS} doesn't. The relation between CH and M0 pitch among all FFETs and CFETs in Fig. 7(e) shows the FFET has better scalability than all the CFETs thanks to the F4ET's most relaxed M0 pitch.

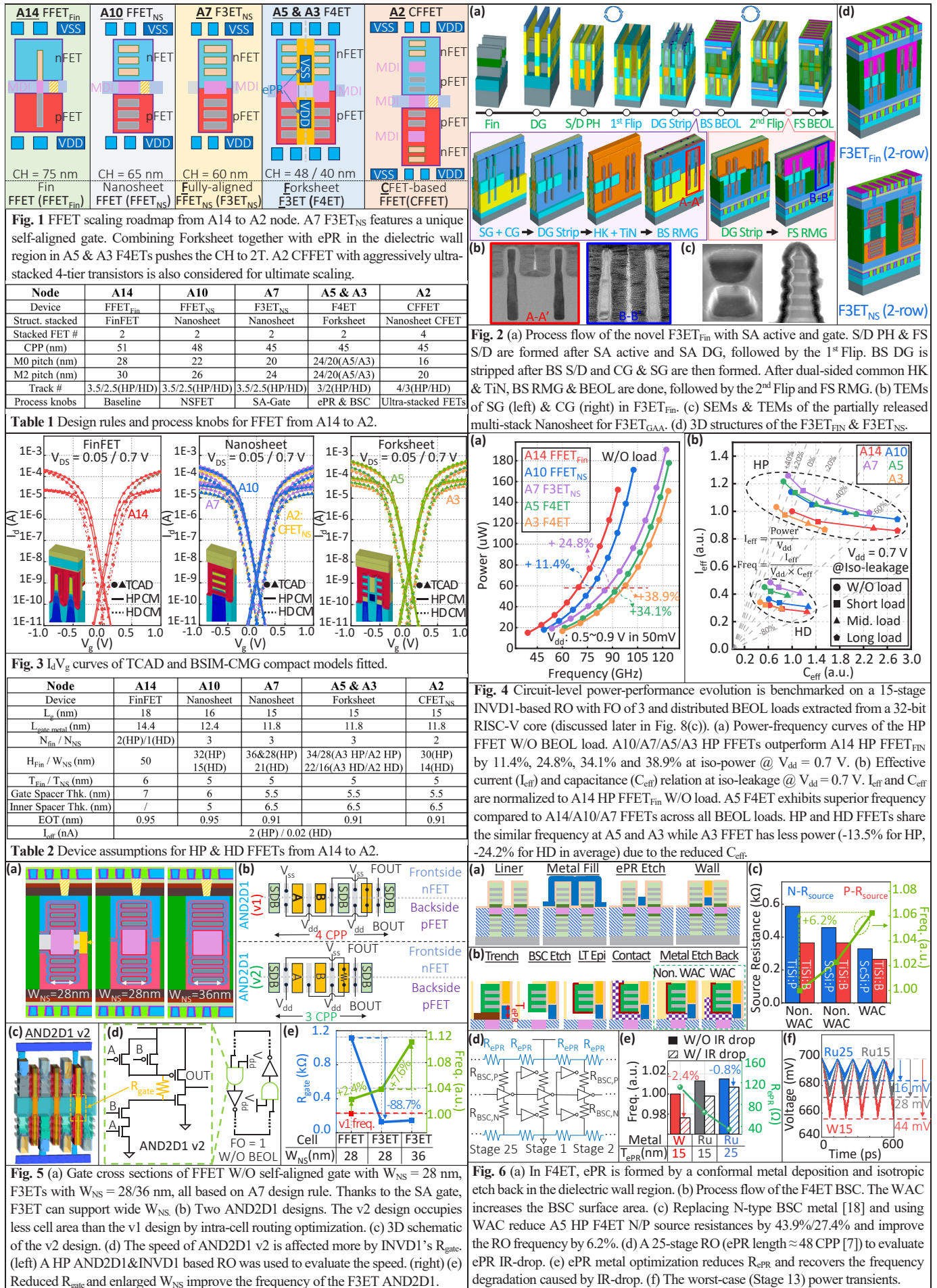
Block-level PPA Scaling: Physical implementation (including dual-sided P&R) and PP evaluation [24] were carried out on 32-bit RISC-V cores based on A14-A5 FFET libraries @ $V_{dd} = 0.7$ V. BEOL design rules are listed in Table 3. Post P&R HP core layouts in Fig. 8(a) shows the continuous area scaling down to A5, consistent with Fig. 8(b). Statistics of BEOL layers in CPs are given in Fig. 8(c). Compared to the A14 HP FFET_{Fin}, the A10, A7 and A5 HP FFET exhibit continuous scaling in performance (max 20% at iso-power) and energy efficiency (max >15% of average EDP), as in the Figs. 9(a-c). Fig. 9(d) shows the A5 HD F4ET has 49.8% smaller core area, 27.9% higher frequency and 13.1% smaller EDP compared to the A14 HD FFET_{Fin}, further validating the PPA benefits.

Ultra-stacked Transistors at A2: As the first-of-a-kind 4-tier transistors with back-to-back-stacked CFETs, the CFFET is an ultimate extension of the FFET technology. As shown in Fig. 10(a), the dual-sided signal & power, the VHV routing scheme [25] and the super vias enable the aggressive intra-cell routing in the CFFET. Nanosheet, instead of Forksheet, is more suitable for the CFFET due to the needs of super S/D vias on each side of the active. The sheet # (N_{NS}) of the CFFET is limited to 2 considering the high AR constraints. A2 HP CFFET's cell area shows ~ 8.4% area gain over A3 HP F4ET, while HD cells are similar (Fig. 10 (b-c)). Unfortunately, INVD2-based RO results show the A2 HP and HD CFFET's frequency at iso-power degrades by 9.6% and 23.1% @ $V_{dd} = 0.7$ V w.r.t the A3 ones respectively (Fig. 10(d-e)) as CFFET has fewer N_{NS} , leading to smaller W_{eff} and I_{eff} while C_{eff} is not reduced (Fig. 10(f)) due to the larger parasitics of the super vias.

SRAM Scaling Down to A2: A SRAM array with 256 rows (WL) and 256 columns (BL) was used for the SRAM analysis. Fig. 11(a) shows the FFET SRAM scaling roadmap towards A2 beyond conventional non-stacked transistors [26-27]. The unique Bipolar SRAM by FFET [8] is a little smaller than the Unipolar SRAM [8] due to the removal of the super vias (Fig. 11(b)). Fig. 11(c) gives the CFFET SRAM, with two stacked CFET SRAMs [28]. Thus, the A2 CFFET SRAM array can be column-folded (256 rows×128 cols) or row-folded (128 rows×256 cols) (Fig. 11 (d)), gaining 50% area at the same memory size. BL & WL RC (Fig. 11(e)) were considered in the worst-case cell simulation, as given in Fig. 12. Bipolar SRAM exhibits smaller write delay than Unipolar SRAM at A14-A5 for the smaller WL C & BL R and smaller read delay at A7-A5 for the reduced BL C. However, the write speed and write margin of Bipolar SRAM degrades greatly at A3 for the significant BL R degradation, which can be recovered for A2 CFFET SRAM with folded row design due to halved BL RC.

Conclusion: With architecture innovations, in this work, the FFET technology further extends into the F3ET with SA gate and the F4ET with smallest 2T design enabled by the ePR. Through comprehensive PPA evaluation, we validated the great scalability of the FFET down to A3 node (better than the CFET w/ BSC). Though A2 CFFET with ultra-stacked 4-tier transistors has some area gains but no obvious PP gains for logic application, it shows great potential to push the SRAM scaling beyond A2.

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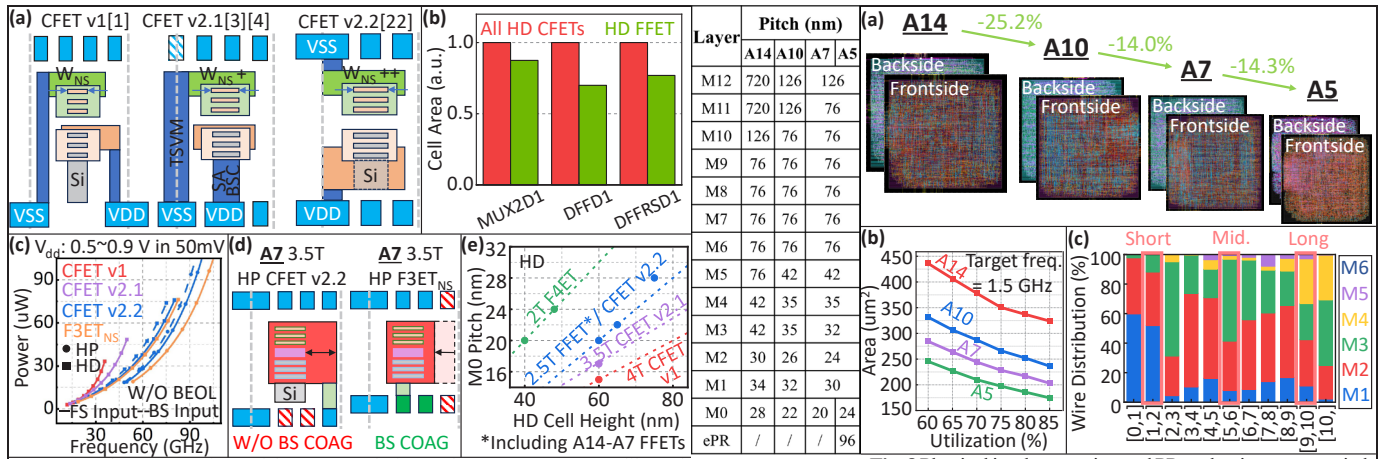


Fig. 7 (a) S/D cross sections of three CFET designs. The CFET v2.2 was calibrated to [22]. (b) The FFET shows area gain over all the CFETs in cells with SG, taking three HD cells as examples. (c) Power-frequency curves comparing F3ET_{NS} and the three CFETs in (a) at A7. F3ET_{NS} shows the best performance due to its largest W_{NS} and smallest parasitics, especially with BS input pin. (d) 3.5T HP CFET v2.2 needs larger gate extension due to the lack of BS COAG, causing larger gate capacitance. (e) M0 pitch vs HD CH. The FFET (F4ET) shows the best area scalability.

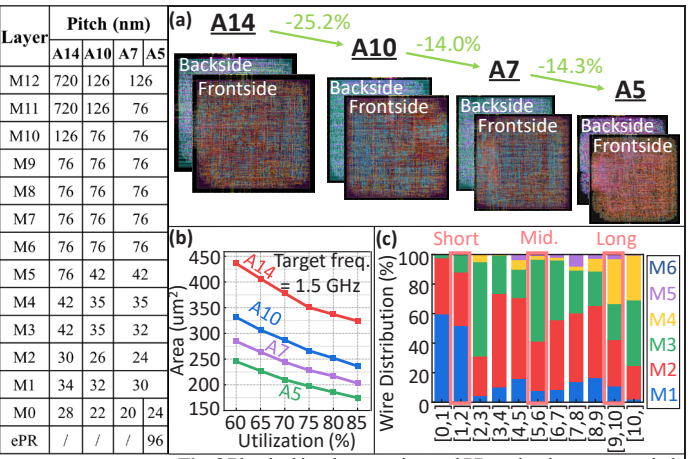


Fig. 8 Physical implementation and PP evaluation were carried out on 32-bit RISC-V cores based on A14-A5 FFET libraries @ $V_{dd} = 0.7$ V (each has > 50 cells), with utilization = 80% and target frequency = 1.5 GHz: (a) Dual-sided HP core layouts. (b) Area vs utilization. (c) BEOL layer distribution for nets in the critical paths. FS and BS metals at the same level (e.g. FS M1 & BS M1) are considered the same.

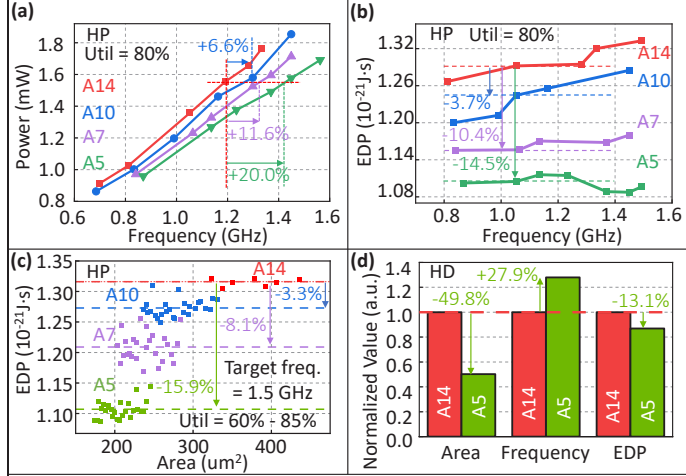


Fig. 9 Block-level PPA results from A14 to A5 FFET RISC-V cores: (a) Power-frequency plot of HP FFET cores at utilization = 80%. (b) EDP vs achieved frequency in HP FFET cores at utilization = 80%. (c) EDP vs core area in HP FFET cores at target frequency = 1.5 GHz and utilization from 60% to 85%. (d) Area, achieved frequency and EDP comparison between the A14 HD FFET_{Fin} and A5 HD FFET at the same target frequency = 800 MHz.

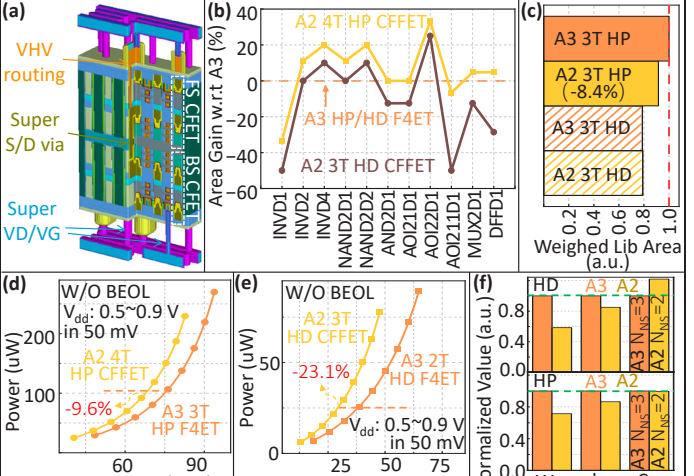


Fig. 10 (a) 3D schematic of the CFET AOI211D2. (b) Standard cell area comparison between A3 F4ET and A2 CFFET. (c) Weighed library area (extracted from RISC-V cores) scaling from A3 to A2. Power-frequency curves of the 15-stage INV2D-based RO comparing A3 F4ET and A2 CFFET in (d) HP and (e) HD cells. (f) W_{eff} , I_{eff} and C_{eff} of the A3 F4ET and the A2 CFFET.

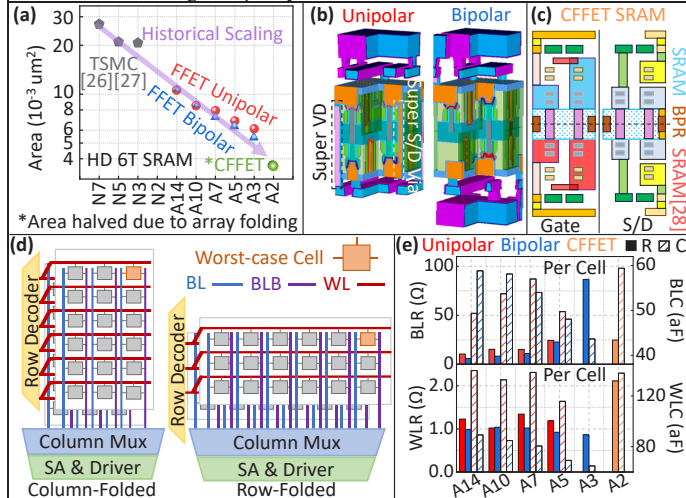


Fig. 11 (a) The SRAM scaling roadmap down to A2. (b) 3D structures of the FFET Unipolar and Bipolar SRAM (smaller due to no super vias). (c) Schematics of the CFFET SRAM in gate and S/D cross sections. BPR is shared by both sides. (d) Two folding schemes for the CFFET SRAM array. (e) WL & BL RC per bit cell at A14-A2. A3 Bipolar SRAM BL R degrades greatly for its narrowest BL.

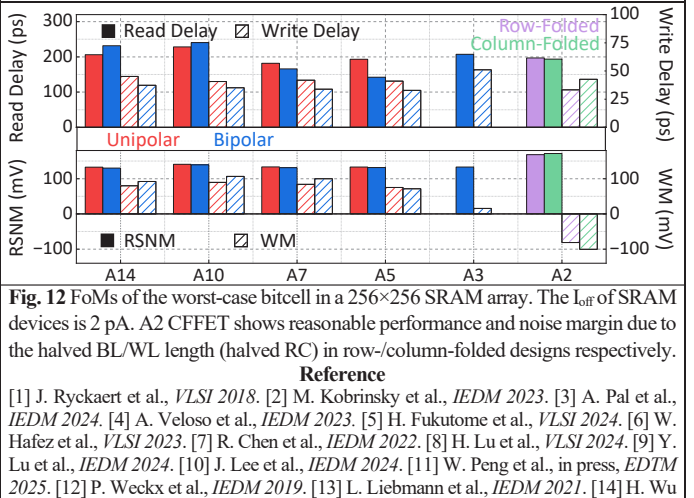


Fig. 12 FoMs of the worst-case bitcell in a 256x256 SRAM array. The I_{on} of SRAM devices is 2 pA. A2 CFFET shows reasonable performance and noise margin due to the halved BL/WL length (halved RC) in row/column-folded designs respectively.